FOR OFFICIAL USE

N5

National Qualifications 2026

Mark

X816/75/01 Computing Science

WEDNESDAY, 20 MAY
9:00 AM – 10:30 AM

X8167501

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth
Day Month Year

Scottish candidate number

Total marks — 80

SECTION 1 — Software design and development, and Computer systems — 55 marks
Attempt ALL questions.

Attempt EITHER Section 2 OR Section 3

SECTION 2 — Database design and development — 25 marks

SECTION 3 — Web design and development — 25 marks

You may use a calculator.
Show all workings.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Do not remove any exam materials. You must leave this booklet on your desk; if you do not, you could lose all the marks for this paper.

Qualifications Scotland
Teisteanasan Alba

X8167501 01

SECTION 1 — SOFTWARE DESIGN AND DEVELOPMENT, AND COMPUTER SYSTEMS — 55 marks

Attempt ALL questions

1. A company wants to reduce its energy consumption.

Describe two methods to reduce the energy consumption of its office computers when in use. **2**

Method 1 _____

_____

Method 2 _____

_____

2. The weekly average screen time use of a tablet is calculated and stored in the variable `screenTime`.

Using a programming language of your choice, write the code to store this weekly average screen time to two decimal places. **2**

3. An online shop uses a firewall.

Describe how a firewall ensures customer data is kept secure. **1**

_____

_____

4. The following code checks if a pupil will receive a merit.

```
...
Line 34   IF peerMentor = TRUE AND attendance > 95 THEN
Line 35       SET merit TO "Yes"
Line 36   END IF
...
```

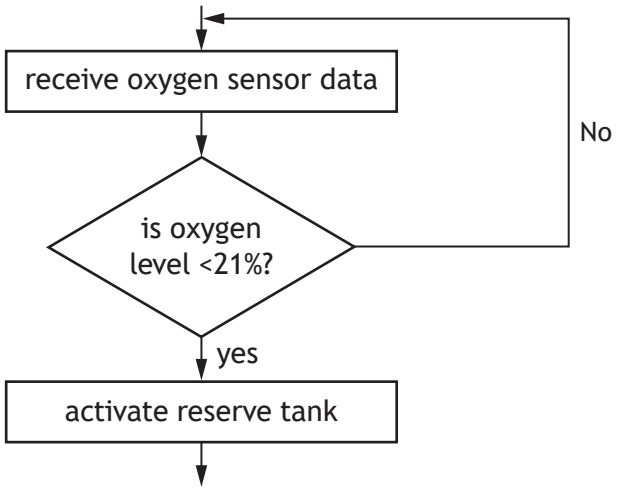
(a) Identify the logical operator in this code. **1**

(b) This code is tested using normal test data.

(i) Describe the purpose of using normal test data when testing code. **1**

(ii) Give an example of normal test data that could be used on line 34. **1**

5. The oxygen level in the International Space Station is monitored using sensors. Part of the design of a program to control the oxygen level is shown below.



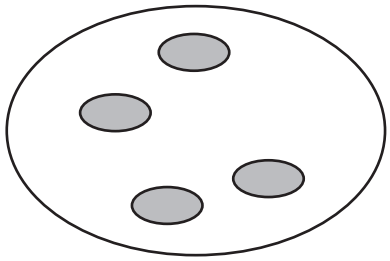
Name the coding construct shown in this design. **1**

[Turn over

6. AquaQuest Explorations measures the depth of a loch. The greatest depth they measured was 114 metres.

(a) Convert the denary number 114 into 8-bit binary. **1**

(b) A map of the loch is shown below.



The map is saved as a vector graphic.

(i) State one object used in this graphic. **1**

(ii) This graphic could also be stored as a bit-mapped graphic.

Describe how a bit-mapped graphic would be stored in a computer system's memory. **2**

7. Explain why the development of a program is an iterative process. **1**

8. SkySprings Trampoline Park is developing a program to allow customers to book group entry to the park.

The program will ask the user how many people are in the group and if each person would like refreshments (Y or N). The program will display the total cost.

(a) Identify two processes for this program. **2**

Process 1 _____

_____

Process 2 _____

_____

(b) The program will use a loop to ask if each person in the group would like refreshments.

State the type of loop that should be used. Explain your answer. **2**

_____

_____

_____

(c) When a booking is made the user must create a password. The password is stored in `userPass`.

Using a programming language of your choice, write the code to ask the user to enter their new password, and calculate the number of characters it contains. **2**

(d) Complete the table below to state which data type should be used for the `cost` and `refreshments` variables. **2**

Variable	Sample data	Data type
cost	27.50	
refreshments	Y	

MARKS DO NOT WRITE IN THIS MARGIN

9. A program is being designed to allow customers to hire paddleboards.



The algorithm for this program is shown below.

1. Initialise variables
2. Ask user for valid number of hours
3. Use number of hours to assign cost for one paddleboard
4. Ask user for number of paddleboards required
5. Ask user for the number of wetsuits required
6. Calculate totalHire
7. Display totalHire

(a) Using a design technique of your choice, refine step 2 to ensure the user enters a valid number of hours. **4**

9. (continued)

(b) Using a design technique of your choice, refine step 6. **3**

(c) State the part of the processor that will temporarily store the calculated total cost of a hire once the program is implemented. **1**

_____

[Turn over

10. Scottish Space Agency is recruiting staff for its space exploration project.

(a) A program is being developed that will allow applicants to complete an electronic application form. They must enter their name, age and the job they want to apply for. Once this information is entered, the interview date for the chosen job is displayed.

Design a user-interface for the:

(i) input for this program **1**

(ii) output for this program. **1**

10. (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(b) There were 84 applicants. The applicants' ages are stored in a data structure called `ages`.

(i) Identify the most suitable data structure and data type for `ages`. **2**

Data structure _____

Data type _____

(ii) Using a programming language of your choice and the data structure `ages`, write the code to calculate and display the average age of applicants. **4**

[Turn over

10. (continued)

(c) The launch speed of a rocket is 7.83 kilometres per second. This value would be stored in a computer system using floating-point representation as shown below:

0.783×10^1

Identify the mantissa and exponent in the above floating-point representation. **2**

Mantissa _____

Exponent _____

11. In an online game, players control a character in a fantasy world.

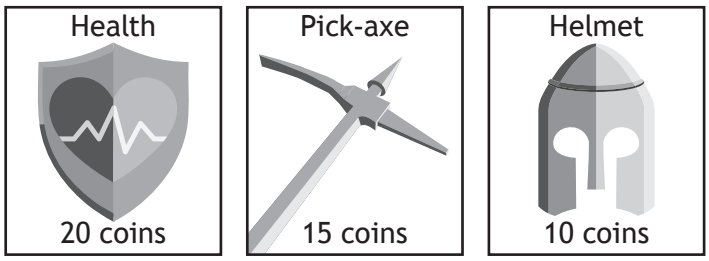
(a) The character's status is determined by their current health and food scores using the following rules:

- 'Feeling fine' — your character has 40 or more food points and 50 or more health points.
- 'Getting tired' — your character has under 40 food points or under 50 health points.
- 'Help' — your character has any other combination of points.

Using a design technique of your choice, design the code to determine the character's status. **4**

MARKS | DO NOT WRITE IN THIS MARGIN

(b) Players collect coins that they can use to buy upgrades.



The number of coins collected and spent by a player are stored in the variables `coinTotal` and `coinsSpent`.

Part of the code for this game is shown below.

```
...
Line 13  <player collects coins>
Line 14  <player spends coins>
Line 15  SET coinTotal TO coinTotal - coinsSpent
Line 16  IF coinTotal < 50 THEN
Line 17      SET bonusCoins TO 100 / coinTotal
Line 18  END IF
Line 19  SEND "You have" TO DISPLAY
Line 20  SEND bonusCoins TO DISPLAY
Line 21  SEND "bonus coins" TO DISPLAY
...
```

A player has collected 10 coins and has bought a helmet. The program stops and produces an error at line 17.

State the type of error that has occurred at line 17 and explain why this has happened. **2**

Type of error _____

Explanation _____

_____

(c) Name the part of the computer architecture that transfers the value of `coinTotal` from main memory to the processor. **1**

_____

(d) Using a programming language of your choice, write the code to combine the outputs in lines 19, 20 and 21 into a single output message that tells the player how many bonus coins they have. **2**

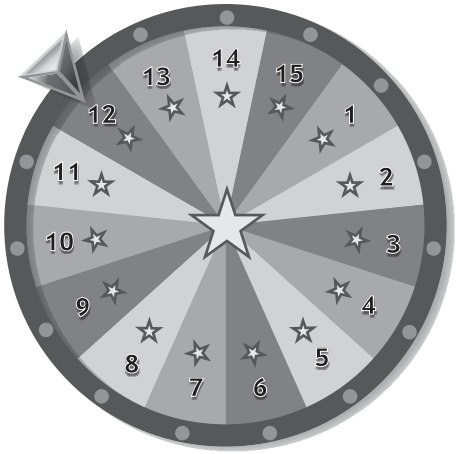
MARKS

DO NOT WRITE IN THIS MARGIN

(e) A programmer commented on the readability of the program.

Explain what is meant by readability. Your answer should refer to one example from lines 15–18 in the code on the previous page. **2**

Every day players are given a free spin of a bonus wheel to win more coins. Players will receive between 1 and 15 coins.



12 bonus coins received

(f) Using a programming language of your choice, write the code to generate and store a value between 1 and 15 in a variable called `bonus`. **3**

(g) During the software development process, the code must be translated.

State the type of translator that should be used when the game is released to the public. **1**

[END OF SECTION 1]

[BLANK PAGE]

DO NOT WRITE ON THIS PAGE

SECTION 2 — DATABASE DESIGN AND DEVELOPMENT — 25 marks

Attempt ALL questions

12. A card collector uses a database to store information on the cards in their collection.

Some of the data currently stored is shown below.

Collection					
cardName	**series**	**expansion**	**setNum**	**cardCode**	**graded**
Night Dog	Little Monster	Sparkling Sea	92477	D5826	True
Alex Kaur	MBA League	Legendary Ballers	16748	F9730	True
Fire Mouse	Little Monster	Evolved Monsters	66438	B7351	False
Unblock	Spells Gathered	Trapped Potential	59351	C1043	True
Sam Speed	MBA League	The Court	30134	E2904	False
...	...	...	...	...	...

The collector no longer collects MBA League cards and wants to remove them from the database.

The following SQL statement is written to remove these records.

```
DELETE FROM Collection
WHERE expansion = "The Court";
```

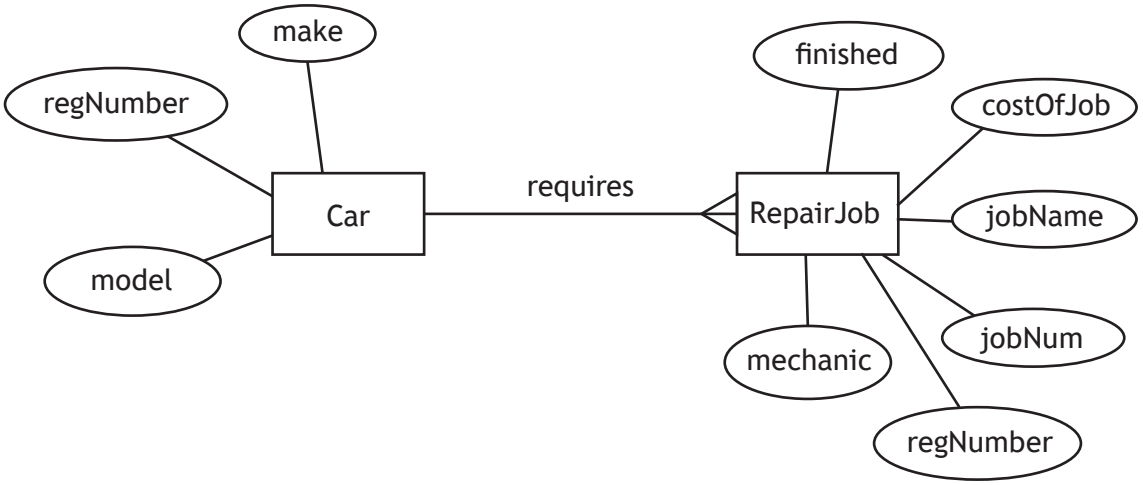
(a) Explain why this SQL statement is not fit for purpose. **1**

(b) Rewrite the SQL statement to remove the records correctly. **1**

[Turn over

13. An entity-relationship diagram was created to design a database for a garage.

Complete the entity-relationship diagram below by identifying the primary and foreign keys. **2**



MARKS | DO NOT WRITE IN THIS MARGIN

14. A school carried out a survey to find out how pupils travel to school. Pupils completed the survey shown below.

School Travel Survey

Please answer the questions below.

Mode of transport

○ Car ○ Bus ○ Walk ○ Cycle ○ Other

Class

Scottish Candidate Number

(This is your 9-digit code)

All results were stored in a database. A sample record is shown below.

scotN	class	transport
132548702	4N1	Bus

Complete the table below to identify the validation for the fields scotN and transport. **2**

Field	Validation
scotN	
transport	

DO NOT WRITE IN THIS MARGIN

15. A library uses a relational database to store the details of books that have been borrowed and the members who have borrowed them.

A sample of the data stored in each table is shown below.

Member			
memberName	**contactNum**	**memNum**	**dateOfBirth**
Jo Thompson	07721454187	B4L04062	25/08/1988
Skyler Parker	07895582319	DB3S3741	01/08/2003
Sam Smart	07378565190	K2S06536	11/09/2010
Casey Kim	01415571239	LO6N6448	16/10/1991
Ellis Chen	07178251143	MN567379	20/06/2012
Aliyah Taylor	01415829631	T2D26905	05/06/2009
Ari Morgan	01418529637	V0D33927	30/01/1993
…	…	…	…

Borrowed					
title	**author**	**genre**	**borrowRef**	**memNum**	**dateBorrowed**
Hunting the Skylark	Harper Bea	Historical	678 0062	B4L04062	09/02/2026
Hunting the Skylark	Harper Bea	Historical	678 4578	DB3S3741	28/12/2025
The Drifter's Diary	J Dorean	Fiction	678 0316	T2D26905	21/04/2026
Year Zero	Brodie Arwen	Horror	678 0454	K2S06536	13/04/2026
Blooded Prince	Bran Harper	Horror	678 0486	T2D26905	14/02/2026
Forgotten Map	H.P.P. Sullen	Adventure	678 0547	V0D33927	03/03/2026
The Silver Masquerade	J. Ronald Fitzgerald	Classic	678 0743	DB3S3741	21/02/2026
Vanity and Valor	Janey Alexander	Classic	678 1563	LO6N6448	19/03/2026
…	…	…	…	…	…

15. (continued)

(a) Explain why the `borrowRef` field is stored as a text data type. **1**

(b) A publishing company asks the library to provide the contact details of all members who have borrowed an adventure book.

(i) Design a query that could be used to generate a list of names and phone numbers of members who have borrowed an adventure book before 15/03/2026 with the most recent date borrowed first. **5**

Field(s)	
Table(s)	
Search criteria	
Sort order	

(ii) A member has complained that they have been contacted by the publishing company. The member tells the library that they have breached UK General Data Protection Regulation (UK GDPR).

Explain why the library has breached UK GDPR. **1**

[Turn over

15. (continued)

(c) A new member has joined the library.

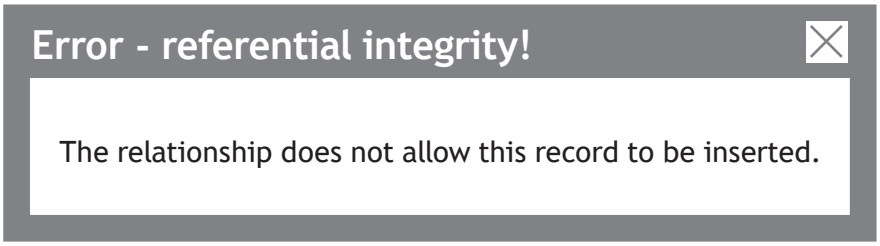
Their details are added to the `Member` table by executing the following SQL statement.

```
INSERT INTO Member (memberName, contactNum, memNum,
dateOfBirth)
VALUES ("Darcy Hughes", "01415831584", "CH0S1162",
"07/07/2003");
```

Once they have a membership, they borrow the 'Classic' book 'Beyond the Veil' by Billy Jameson on 28 April 2026.

Write an SQL statement that will add this record to the `Borrowed` table, using the `borrowRef` '678 0061'. **2**

(d) The librarian tries to add a new record to the `Borrowed` table, but the error message below is displayed.



Explain why referential integrity has been broken. **1**

[Turn over for next question

DO NOT WRITE ON THIS PAGE

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

16. A restaurant manager keeps information about staff and bookings in a relational database. Sample data from both tables is shown below.

Staff		
employeeID	**staffName**	**contractHours**
99	Steven Miller	15
213	Mary Joyce	35
102	Lily Stewart	28
…	…	…

Booking						
bookingRef	**tableNum**	**bookingName**	**time**	**date**	**numGuests**	**employeeID**
A546	5	Smith	18:45	03/04/2026	5	213
A785	10	Howie	15:00	05/04/2026	2	102
B124	2	Barr	19:00	01/04/2026	9	99
F368	6	Grant	12:00	04/04/2026	3	213
…	…	…	…	…	…	…

(a) The manager would like a list of staff names and number of guests for all bookings after 21:00 with five or more guests.

Complete the SQL statement below that would produce this output. **5**

```
SELECT _____

FROM _____

WHERE _____

_____

_____
```

16. (continued)

(b) Mary Joyce is not able to come to work on 4 June 2026. Steven Miller will now serve the bookings that Mary had been allocated.

Complete the SQL Statement below to update the database. **4**

```
UPDATE _____

SET _____

WHERE _____

_____

_____
```

[END OF SECTION 2]

[Turn over

[BLANK PAGE]

DO NOT WRITE ON THIS PAGE

SECTION 3 — WEB DESIGN AND DEVELOPMENT — 25 marks

Attempt ALL questions

17. The following HTML code is added to each page of a website to link to an external stylesheet.

```
<link rel="stylesheet" type="text/css" href="style.css">
```

(a) Describe the purpose of a stylesheet. **1**

(b) Explain why an external stylesheet is used. **1**

[Turn over

18. Part of a STEM careers website is shown below.

https://www.stemcareers.co.uk

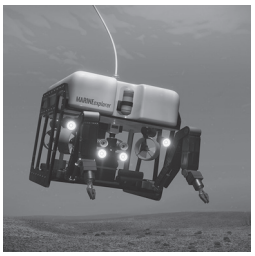
STEM careers

Water covers about 71% of the Earth's surface, so there's a lot to discover at the bottom of the sea and lakes!

Underwater archaeologist

Underwater archaeology is a bit trickier than normal archaeology. It involves techniques such as surveying sites with sonar, depth gauges and ROVs (Remotely Operated Vehicles).

It's an exciting field, with a great deal of depth.



The headings 'STEM careers' and 'Underwater archaeologist' have been styled using the following CSS rule.

```
.headings{text-align: center; font-size: 26px;}
```

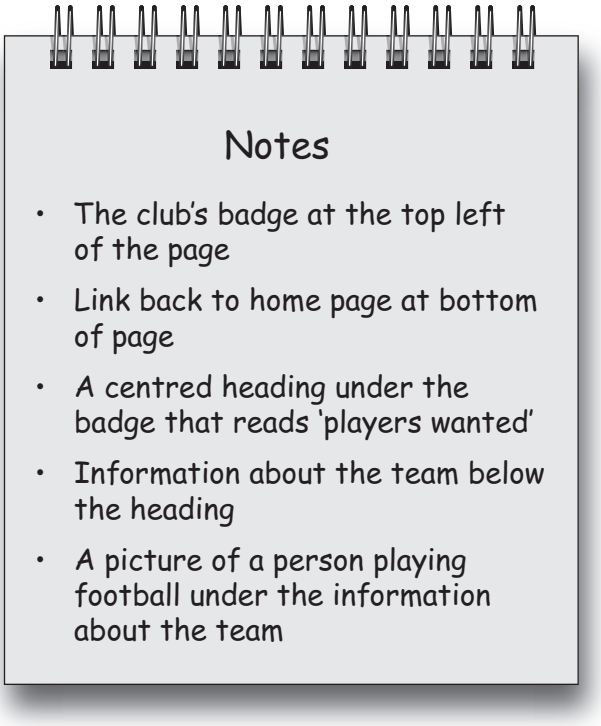
Explain why a class has been used. **1**

_____

_____

19. A football team would like to add a page to their website to advertise for new players.

After meeting with the team's coach, the requirements were noted as shown below.



Complete the wireframe below to design this web page. **3**



20. Gary has a website for his garden centre.

(a) The home page tab displays the name of the web page as 'home.html' as shown below.



This should be displayed as 'Gary's garden centre'.

Write the code to implement this. **1**

The HTML and the CSS code for the home page is shown below.

HTML code

```
…
<body>
<h1 class="style1"> GGC </h1>
<h2 id="style2"> Plants </h2>
<p class ="style1"> The best plants in town </p>
<img src="lily.jpg" />
<p> Here are some of our best sellers </p>
<ul>
    <li> Lily </li>
    <li> Rose </li>
    <li> Cactus </li>
    <li> Jade </li>
</ul>
</body>
</html>
```

CSS code

```
.style1{text-align:center;}

p{text-align:left;}

#style2{text-align:right;}
```

20. (continued)

(b) Draw how the home page will look when viewed in a browser.
Some of the content has already been added. **3**

https://www.garysgardencentre.co.uk

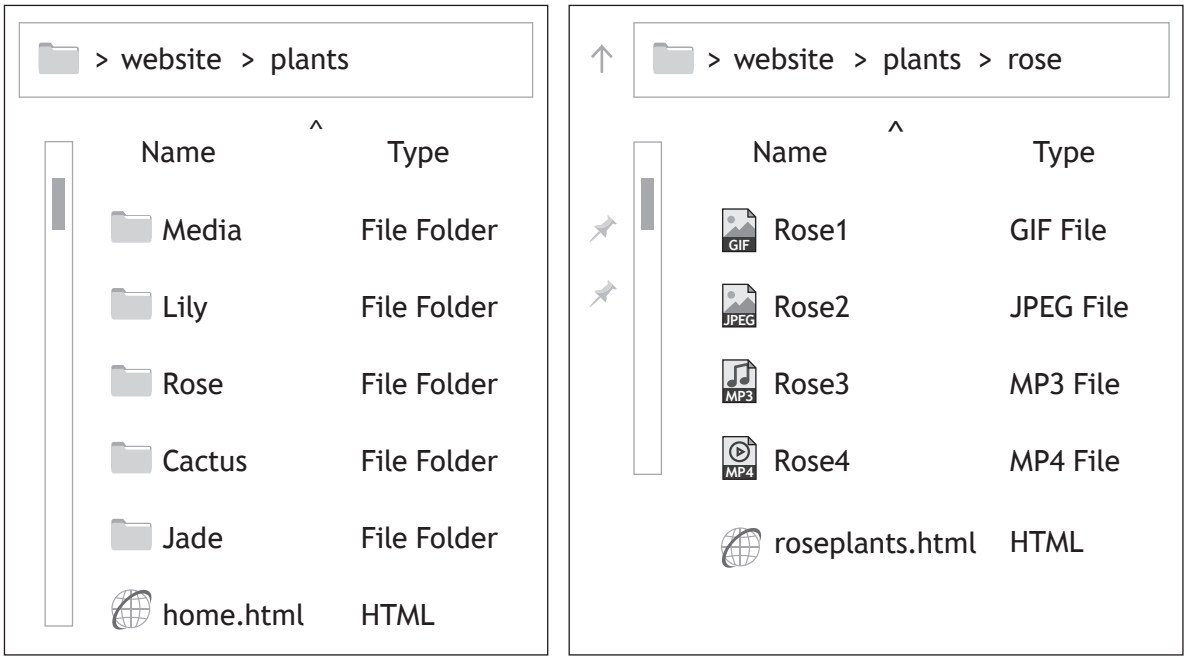
The best plants in town



Here are some of our best sellers

20. (continued)

(c) All the files for the website are stored in a folder called 'plants' as shown below.



(i) Gary wants to add links to his best-selling plants list.
Complete the following code to add the link to the 'Rose' page. **3**

```
<li>


</li>
```

(ii) Gary wants to add the animated picture of a rose to the 'Rose' page.
Write the code to implement this. **2**

20. (continued)

(d) The 'Cactus' page includes a feature that changes an image of a young cactus plant, to an image of a fully grown cactus plant, when the cursor is placed on the image.

(i) State the programming language used to implement this feature. **1**

_____

(ii) State the type of event that is used. **1**

_____

(e) Gary would like to use pictures of customers' gardens on the website.

State the name of the legislation that applies to the use of these pictures. **1**

_____

[Turn over

21. A restaurant has a website.

The HTML and CSS code for the home page is shown below.

HTML code

```
…
<body>
<div>
<h1> Bella Restaurant </h1>
<h2> The best restaurant in town </h2>
<p> Welcome to our restaurant </p>
</div>
<div id ="menu">
<h2> Italian food </h2>
<p> Pasta</p>
<p> Pizza</p>
</div>
…
```

CSS code

```
div{background-color:green;text-align:center;}
p{font-size:18px;font-family:verdana;}
#menu{background-color:red;text-align:right;}
body{background-color:blue;font-family:arial;
font-size:20px;}
```

(a) Complete the table below to identify how this code would display the 'Italian food' and 'Welcome to our restaurant' text in a browser. **4**

	'Italian food'	**'Welcome to our restaurant'**
Font		
Text size		
Background colour		
Text alignment		

(b) The restaurant wants to add a link to one of their supplier's website.

'www.lowdensuppliers.com'

State the type of addressing required. **1**

_____

21. (continued)

(c) The restaurant has added an audio file to the website using the following code.

```
<audio controls>
   <source src="food.wav" type="audio/wav">
</audio>
```

The file is converted to an MP3 file and replaces the audio file on the website.

Describe one benefit and one drawback of this change to the website user. **2**

Benefit _____

_____

Drawback _____

_____

[END OF SECTION 3]

[END OF QUESTION PAPER]

DO NOT WRITE IN THIS MARGIN

ADDITIONAL SPACE FOR ANSWERS

Looking for more resources? Visit scotpapers.com – Scotland's #1 Past Paper Archive

DO NOT WRITE IN THIS MARGIN

ADDITIONAL SPACE FOR ANSWERS

[BLANK PAGE]

DO NOT WRITE ON THIS PAGE

Acknowledgement of copyright

Question 18 Vismar UK/shutterstock.com

Question 20 (b) Adam Yee/shutterstock.com